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Enantioselective Alkenylation of Aldehydes with Protected Propargylic Alcohols in the Presence of a Crown Ether as an Additive.

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Abstract. The enantioselective alkenylation of aldehydes with protected propargylic alcohols catalyzed by chiral amino thiol has been developed. The best results were obtained for the hydrozirconation-transmetallation to zinc protocol. The key element of the method is the use of crown ethers as an additive. The enantioselectivity of the reaction performed in the presence of the crown ether was substantially higher than in the absence of this additive and the enantiomer ratio reached 98:2.

Keywords: aldehydes; allylic alcohols; asymmetric catalysis; crown ethers; enantioselectivity

Enantioenriched allylic alcohols are extremely valuable and versatile starting materials for synthesis. The presence of a double bond and a hydroxyl group allows for various manipulations leading to synthetically useful building blocks, available by such double bond functionalization reaction as the Simons-Smith cyclopropanation,[1] epoxidation,[2] dihydroxylation,[3] and many others. Therefore, efficient methods for the synthesis of enantioenriched allylic alcohols are of great value.[4] One of most valuable methods is a 1, 2-nucleophilic addition of organometallic reagent to a carbonyl group of aldehydes and ketones. The best results are obtained when alkylalkenylzinc compounds are used. This approach was first developed by Oppolzer and Radinov,[5] and then highly improved by Walsh,[6] and consists of the three subsequent reactions – hydroboration of an alkyne, transmetallation of a resulting vinylborane with dialkylzinc and the addition of alkylalkenylzinc reagent to the carbonyl compound. The second approach evolved from Wipf's work on hydrozirconation reaction. In this case, the first step is the hydrozirconation reaction of an alkyne with a Schwartz's reagent, while the next two steps remain the same.[7] This methodology was thoroughly studied and variously substituted alkynes were used yielding respective allylic alcohols with excellent chemical yields and enantioselectivities.[7] However, while several acetylenes bearing terminal hydroxyl group were used, no successful results were reported for propargylic alcohols. This three-carbon acetylenic alcohol employed for the enantioselective alkenylation can yield highly valuable allylic alcohol bearing two hydroxyl groups, one primary and one connected to the stereogenic carbon (Fig 1).

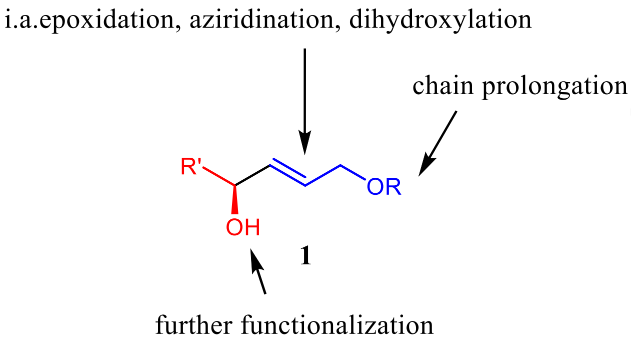

Figure 1. Synthetic possibilities given by allylic alcohols derived by enantioselective alkenylation of aldehydes with propargylic alcohols.

This type of allylic alcohol plays an important role in our ongoing project on Achmatowicz rearrangement (Fig. 1, R=furyl) and its application in the synthesis of polyhydroxylated natural products. According to our best knowledge, there is only a single publication dealing with the problem. Bräse and co-workers attempted to use Oppolzer's methodology to synthesize **1**, however, no expected product was observed, supposedly due to competitive side reaction.

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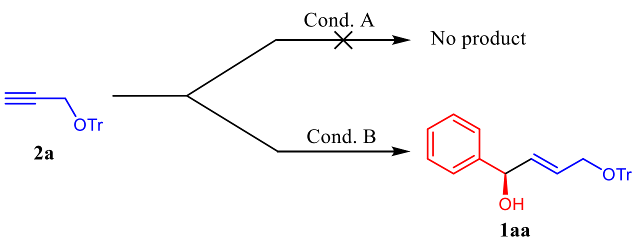
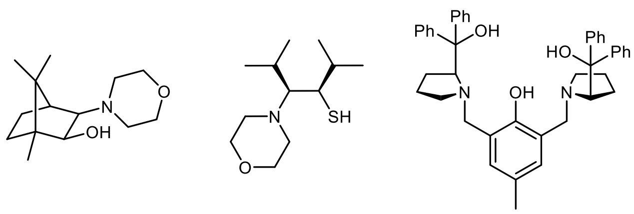
Under Wipf's conditions, the reaction proceeded well, albeit with negligible enantioselectivity.[8] Hence, we decided to investigate this reaction more closely.
At the beginning of our study, we decided to perform the reaction under Oppolzer's condition using trityl-protected propargylic alcohol. We wanted to assure ourselves, that the lack of positive result is not ligand, but conditions dependent. Instead of paracyclophane ligand used by Bräse, we used MIB **L1** (Fig.2), which efficacy in alkenylation reactions has been proved several times. Not surprisingly, we failed to obtain the desired alkenylation product, either. Therefore we decided to thoroughly investigate this reaction under Wipf's conditions (Scheme 1).



Scheme 1. Alkenylation of benzaldehyde. Conditions A: a) Cy_2BH, a) Et_2Zn, c) PhCHO (**3a**). Conditions B: a) $Cp_2Zr{\cdot}HCl$, b) Et_2Zn, c) PhCHO (**3a**).

The reaction was performed in the presence of 4 mol% of MIB and yielded expected products with very good yield. However, the isolated products were racemic.

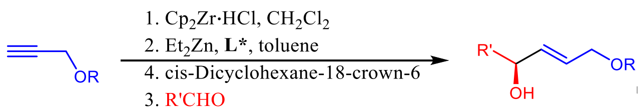


L1 **L2** ***(S,S)*-L3**

Figure 2. Ligands used in this study.

We turned our attention to another excellent ligand efficient in alkenylation reactions, namely to amino thiol **L2** (Fig.2).[9, 10] Thiols, due to high affinity to zinc are more efficient ligands than respective amino alcohols.[11-15] The result of the first performed reaction was very promising (Table 1, entry 1). Then we started to investigate the influence of the solvent on the reaction. The best results were obtained in the anhydrous ethyl ether (Table 1, entry 2). The influence of the rate of addition of the aldehyde was also checked. The highest chemical yield and enantioselectivity was obtained, when benzaldehyde was added at the rate of 0.25 mmol/60 sec. (Table 1, entry 5). Slower addition, however, did not improve neither the yield nor the selectivity. Then we investigated the influence of two other parameters of the reaction: the amount of chiral ligand and temperature of the aldehyde's addition. The best results were found for 25 mol% of **L2** and the addition temperature -10°C (Table 1, entries 6-11). The combination of both factors led to so far the best result: – 89% chemical yield and 90.4:9.6 er (Table 1, entry 10).

Table 1. Reaction conditions development.[a]



Entry	Solvent	L* (mol%)	Temp. [°C]	Yield [%]	Er[b]
1	DCM	**L2** (15)	-30	73	67.3 : 32.7[c]
2	Et_2O	**L2** (15)	-30	31	76.7 : 23.3[c]
3	THF	**L2** (15)	-30	35	72.5 : 27.5[c]
4	Toluene	**L2** (15)	-30	51	75.2 : 24.8[c]
5	Et_2O	**L2** (15)	-30	88	86.2 : 13.8[d]
6	Et_2O	**L2** (20)	-30	85	80.5 : 19.5[d]
7	Et_2O	**L2** (25)	-30	84	89.0 : 11.0[d]
8	Et_2O	**L2** (30)	-30	79	76.4 : 23.6[d]
9	Et_2O	**L2** (15)	-20	88	85.6 : 14.5[d]
10	Et_2O	**L2** (15)	-10	89	90.4 : 9.6[d]
11	Et_2O	**L2** (15)	0	84	88.5 : 11.5[d]
12	Et_2O	**L3** (10)	-30	49	31.5 : 68.5 [d]
13	Et_2O	**L3** (20)	-30	46	28 :72[d]
14	Et_2O	**L3** (30)	-30	52	13.5 : 86.5 [d]
15	Et_2O	**L3** (40)	-30	36	9.5 : 90.5 [d]
16	Et_2O	**L3** (60)	-30	-	- [d]

[a] Reactions were run on a 0.25 mol scale. [b] Determined by HPLC with a chiral stationary phase. [c] RCHO addition rate - 0.25 mmol/10 s. [d] RCHO addition rate - 0.25 mmol/60 s.

At this stage we noticed that there is no much room left for the improvement of selectivity by variation of the reaction conditions.
One of the most efficient ligand for the addition organozinc compounds is Trost's Prophenol **L3**.[7] Commercially available in both enantiomeric forms would be an ideal ligand, if working. We decided to test the performance of this ligand (Table 1, entries 12-16).
We were very pleased to report that amino alcohol ligand was working with quite a reasonable enantioselectivity, comparable with that of **L2**. However, this level of selectivity was achieved with 40 mol% of **L3** (Table 1, entry 15), so the mass of ligand was more than two times higher than the mass of benzaldehyde used as a substrate. Moreover, the

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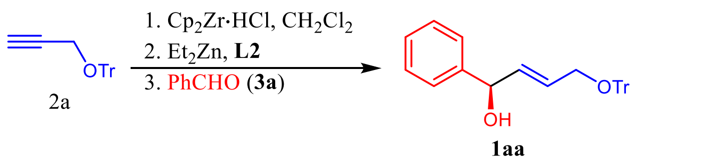
chemical yield was decreasing in the presence of higher amounts of Prophenol. When 60 mol% was used, no expected reaction product could be detected (Table 1, entry 16).
So it seemed that we had entered cul-de-sac. The only obvious reason of the lower enantioselectivity in the reaction catalyzed by the otherwise excellent ligand is the presence of a second product of transmetallation reaction. Apart from the desirable alkylalkenylzinc compound, the zirconocene byproducts are obtained in this reaction. These Lewis acidic compounds are the potent catalyst of the addition of dialkylzincs to aldehydes. It has been experimentally proved by Wipf and co-workers that in the case of alkenylation reaction the zirconocene byproducts promote achiral pathway leading to expected, albeit racemic product with the reaction rate comparable to that performed in the presence of some chiral ligands.[16] At that moment he was unable to suppress this background reaction. The attempts to inhibit achiral pathways in the enantioselective synthesis have been several times described in the literature, with the positive results being reported when 10% of DiMPEG was used.[17-21] The role of (poly)ethylene glycol is the coordination of achiral Lewis acidic byproducts. We decided to use DiMPEG as achiral pathway inhibitor; however, the result was disappointing as the enantioselectivity of the reaction remained on the low level (er 74:26) and the chemical yield dropped to 27%. Therefore we tried to find the another method allowing for the strong complexation of the zirconium byproduct, which, in theory at least, would deactivate it as a Lewis acidic catalyst. Among the most powerful metal complexing agents, crown-ethers plays the leading role. They are used for an extraction of zirconium species,[22, 23] however, to our best knowledge, they were never used as achiral catalysis inhibitors in the enantioselective synthesis. Hence, we decided to try to stop unwanted background reaction by addition of crown-ether which coordination ability should be better than in the case of DiMPEG. After transmetallation step 1.2 equiv. of *cis*-dicyclohexane-18-crown-6 ether was added and the reaction was conducted under the best conditions described above. The result of the first reaction was disappointing; the enantiomeric excess was only slightly better than that obtained in the absence of crown ether (Table 2, entry 1). We decided to change the solvent back to toluene and tried again. We were extremely satisfied with the result of that reaction; the alkenylation product was obtained with excellent er – 97.6:2.4 (Table 2, entry 2). Further optimization (Table 2, entries 3 - 9) allowed for the establishing of the best conditions for the reaction (Table 2, entry 10).

Table 2. Catalytic asymmetric addition in the presence of chiral ligand **L2** and *cis*-dicyclohexane-18-crown-6 ether.[a]



Entry	Solvent	L* mol %	Crown-ether [equiv]	Temp [°C]	Yield [%]	Er[b]
1	Et_2O	25	1.2	-10	60	89.6 : 10.4
2	Toluene	25	1.2	-10	67	97.6 : 2.4
3	Toluene	25	1.2	0	74	97.9 : 2.1
4	Toluene	25	1.2	RT	59	96.8 : 3.2
5	Toluene	25	0.6	-10	77	97.8 : 2.2
6	Toluene	25	0.12	-10	67	94.0 : 6.0
7	Toluene	15	0.6	0	57	95.6 : 4.4
8	Toluene	5	0.6	0	58	93.2 : 6.8
9	Toluene	25	0.6	0	75	97.2 : 2.8[c]
10	Toluene	25	0.6	0	80	98.2 : 1.8

[a] Reactions were run on a 0.25 mol scale. [b] Determined by HPLC with a chiral stationary phase. [c] 18-crown-6 ether was used.

The reaction in the presence of the cheaper 18-crown-6 ether is only slightly less selective. So the use of this less expensive additive can be considered, especially for the larger scale reactions. Next, having established the best conditions for the addition reaction, we tested the scope of the reaction with various alkynes and aldehydes as a substrate. The results are summarized at Figure 3.

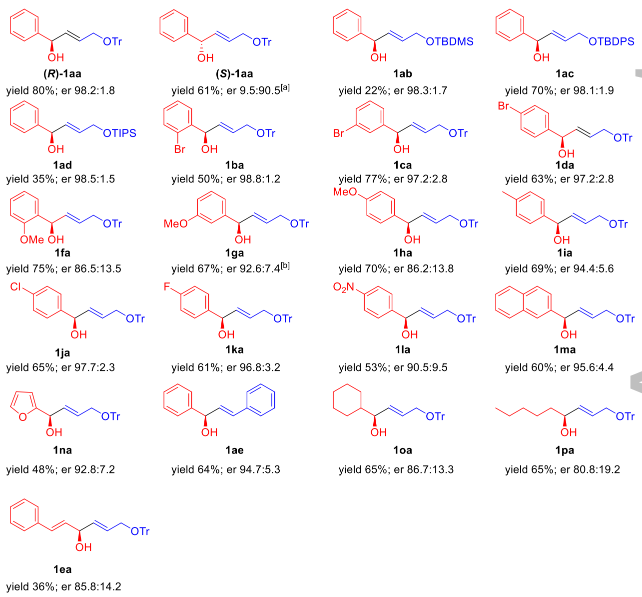


Figure 3. Substrate scope of the asymmetric alkenylation in the presence of chiral ligand **L2** and *cis*-dicyclohexane-18-crown-6 ether. Reactions were run on a 0.25 mol scale. Er was determined by HPLC with a chiral stationary phase.

[a] Compound (*S*)-**1aa** was obtained in the presence of (*S*, *S*)-**L3**. [b] Ee was approximated. We were unable to achieve the complete separation of two peaks (*vide* Supporting Informations)

The highest enantiomeric ratio was observed for aromatic aldehydes, particularly benzaldehydes substituted with electron-withdrawing substituents; the position of the substituent in the phenyl ring did not influence the enantioselectivity. The reactions with aliphatic, cycloaliphatic and α, β – unsaturated aldehydes were less selective. The very high enantioselectivity accompanied by good yield was achieved in the reaction of *tert*-butyldiphenylsilyl-protected alkyne. This is promising in the context of possible applications of the reaction for the synthesis of natural products. The product of the addition of phenylacetylene to benzaldehyde was compared with literature data and its absolute configuration was assigned as *R*. We assumed that the addition of protected propargylic alcohols went according to the same model and the newly formed stereogenic centers had the same spatial arrangements. The products with *S* absolute configuration can be obtained in the reactions catalyzed by **L2**, or better using *ent*-**L3** synthesized from commercially available D-valine.
In summary, we have developed the first highly enantioselective method for the alkenylation of aldehydes with protected propargylic alcohols. The reaction is catalyzed by the amino thiol catalyst in the presence of *cis*-dicyclohexane-18-crown-6 and 18-crown-6 ethers. The crown ether was used as the efficient additive improving asymmetric process. The role of the crown ether requires further studies. The findings open up new avenues to the development of more selective methods in asymmetric synthesis.

Experimental Section

General procedure for the alkenylation reaction

To the suspension of Schwartz's reagent $Cp_2Zr{\cdot}HCl$ (77 mg, 0.3 mmol) in dry CH_2Cl_2 (1 mL) alkyne (0.3 mmol) was added and the mixture was stirred for 10 min at room temperature. Then the solvent was carefully evaporated under vacuum and the residue was dissolved in dry toluene (1 mL). The solution was cooled down to -65°C and 1.1M solution of Et_2Zn in toluene (0.27 mL, 0.3 mmol) was added dropwise. The reaction mixture was stirred for 5 min and the 0.25 M solution of ligand **L2** in toluene (0.25 mL, 14 mg. 0.0625 mmol) was added. The reaction mixture was then warmed to -10°C over 1 hour, *cis*-dicyclohexano-18-crown-6 ether (56 mg, 0.15 mmol) was added and the reaction mixture was warmed to 0°C. Next, the solution of 0.25 mmol of an aldehyde in 0.2 mL of dry toluene was added dropwise over 60 seconds and the reaction was kept at this temperature overnight. After the TLC indicated completion of the reaction, saturated solution of $NaHCO_3$ (10 mL) was added, the layers were separated and the aqueous layer was extracted with Et_2O (3x15 mL). Combined extracts were dried over anhydrous $MgSO_4$, solids were filtered off and the solvents were evaporated. The residue was purified by flash chromatography on silica gel (hexane : ethyl acetate 8:2) to give respective allylic alcohol.

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References

[1] H. J. Kim, P. J. Walsh, *J. Phys. Org. Chem.* **2012**, *25*, 933-938.

[2] A. E. Lurain, P. J. Carroll, P. J. Walsh, *J. Org. Chem*. **2005**, *70*, 1262-1268.

[3] T. J. Donohoe, P. R. Moore, M. J. Waring, N. J. Newcombe, *Tetrahedron Lett*. **1997**, *38*, 5027-5030.

[4] For the review see: A. Lumbroso, M. L. Cooke, B. Breit, *Angew. Chem. Int. Ed.* **2013**, *52*, 1890-1932.

[5] W. Oppolzer, R. N. Radinov, *Helv. Chim. Acta* **1992**, *75*, 170-173.

[6] Y. K. Chen, A. E. Lurain, P. J. Walsh, *J. Am. Chem. Soc.* **2002**, *124*,12225-12231.

[7] For the comprehensive review on the dialkylzinc mediated additions to carbonyl groups, including alkenylation reaction see: T. Bauer, *Coor. Chem. Rev.* **2015**, *299*, 83-150.

[8] F. Lauterwasser, J. Gall, S. Höfener, S. Bräse, *Adv. Synth. Catal.* **2006**, *348*, 2068-2074.

[9] S.-L. Tseng, T.-K. Yang, *Tetrahedron: Asymmetry* **2005**, *16*, 773-782.

[10] P. Wipf, N. Jayasuriya, S. Ribe, *Chirality* **2003**, *15*, 208-212.

[11] J. Kang, D. S. Kim, J. L. Kim, *Synlett* **1994**, 842-844.

[12] J. Kang, J. B. Kim, J. W. Kim, D. Lee, *J. Chem. Soc. Perkin Trans. 2* **1997**, 189-194.

[13] S. Subirats, C. Jimeno, M. A. Pericas, *Tetrahedron: Asymmetry* **2009**, *20*, 1413-1418.

[14] H.-L. Wu, P.-Y. Wu, Y.-N. Cheng, B.-J. Uang, *Tetrahedron* **2016**, *72*, 2656-2665.

[15] P. Wipf, N. Jayasuriya, *Chirality* **2008**, *20*, 425-430.

[16] P. Wipf, S. Ribe, *J. Org. Chem.* **1998**, *63*, 6454-6455.

[17] J. Zhong, H. Guo, M. Wang, M. Yin, M. Wang, *Tetrahedron: Asymmetry* **2007**, *18*, 734-741.

[18] J. Zhong, S. Hou, Q. Bian, M. Yin, R. Na, B. Zheng, Z. Li, S. Liu, M. Wang, *Chem. Eur. J.* **2009**, *15*, 3069-3071.

[19] S. Dahmen, *Org. Lett*. **2004**, *6* , 2113-2116.

[20] C. Bolm, J. Rudolph, *J. Am. Chem. Soc.* **2002**, *124*, 14850-14851.

[21] J. Rudolph, N. Hermanns, C. Bolm, *J. Org. Chem*. **2004**, *69*, 3997-4000.

10.1002/adsc.201900363

[22] N. V. Deorkar, , S. M. Khopkar, *Anal. Chim. Acta*, **1991**, *245*, 27-33.

[23] Y. K. Agrawal, S. Sudhakar, *Sep. Purif. Technol.*, **2002**, *27*, 111-119.

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